

Pre-Fix vs. Post-Fix `diffThreshold` Sensitivity

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Compares binary 0f782fd (tag `binary-v0-buggy`, original) vs. d5bf30c (tag `binary-v1-bugfix`, post-fix).

Resumen

This report repeats the `diffThreshold` sensitivity sweep of `06-diffT-sweep.tex` on the original (buggy) binary (`examine_rewrite` branch) as a sanity check, and compares the two runs side by side. The methodology is identical in every other respect: five threshold values $\{0.05, 0.1, 0.2, 0.3, 0.5\}$ across three configurations (`CPU12`, `GPU`, `GPU+11CPU`), three reps each, plus one `quiet=0` rep per `diffT` of `GPU+11CPU` for max-region data. The headline finding is that *the optimal diffT is the same on both branches* — the hybrid sweet spot sits at `diffT` = 0.1 with means within 0.1 s of each other (52.85 s pre-fix, 52.80 s post-fix). However, the bug fix’s wall-time delta is strongly *config-dependent*: **CPU-only runs ~1% faster post-fix at every diffT**, while **GPU-only runs ~1% slower post-fix at every diffT**, and **the hybrid is essentially neutral** (within $\pm 0.5\%$) across the tighter thresholds, becoming +1.8% slower only at the legacy default of 0.5. The bug-fix overhead is real but is exactly cancelled by improved load balance on configurations where there are multiple CPU workers to absorb the extra leaves; for the GPU-only case (one worker, no load-balance benefit available) the overhead shows up cleanly. The 960×540 , ~15 s worst-case CPU region remains immovable on both branches at every threshold, confirming it is a heuristic problem and neither a tracking-bug artefact nor a threshold-tuning issue.

1. Setup

The binary measured in this report is the one on the `examine_rewrite` branch (the original code, with the `else if min/max` reduction described in `03-bug-analysis.tex`); the binary measured in `06-diffT-sweep.tex` was the `examine_minmax_bugfix` branch. Everything else is identical:

- **Hardware and workload:** i7-9750H + GTX 1660 Ti Max-Q, 100 frames at 1920×1080 , default `pixelSizeThresh` = 32,768.
- **Threshold sweep:** $\{0.05, 0.1, 0.2, 0.3, 0.5\}$. The 0.5 baseline for each branch is reused from the corresponding Fig 11.15 sweep — `experiments/02-fig1115-prefix/` for pre-fix and `experiments/05-fig1115-postfix/` for post-fix.
- **Configurations and reps:** `CPU12`, `GPU`, `GPU+11CPU`; 3 reps each.
- **Max-region pass:** 5 single-rep `quiet=0` runs per branch (one per `diffT`) on `GPU+11CPU` to capture per-region times.

2. Side-By-Side Results

2.1. Wall-Clock Times

Tabla 1: Mean wall-clock time (s) per config and `diffT`, 3 reps each. Δ is post-fix minus pre-fix. Negative Δ means the bug fix *helped* this configuration; positive means it *hurt*.

Config / <code>diffT</code>	0.05	0.10	0.20	0.30	0.50
CPU12 pre-fix	163.12	163.90	163.88	163.58	163.47
CPU12 post-fix	161.47	161.96	162.59	162.24	162.76
Δ (%)	-1.0%	-1.2%	-0.8%	-0.8%	-0.4%
GPU pre-fix	77.27	77.18	77.14	77.14	76.54
GPU post-fix	77.60	77.81	77.82	77.83	78.03
Δ (%)	+0.4%	+0.8%	+0.9%	+0.9%	+1.9%
GPU+11CPU pre-fix	53.24	52.85	53.19	53.30	53.62
GPU+11CPU post-fix	53.45	52.80	52.90	53.21	54.61
Δ (%)	+0.4%	-0.1%	-0.5%	-0.2%	+1.8%

Tabla 2: Population standard deviations (s) per config and `diffT`, 3 reps each.

Config / <code>diffT</code>	0.05	0.10	0.20	0.30	0.50
CPU12 pre/post	1.85 / 0.47	0.45 / 0.71	0.53 / 0.32	0.25 / 0.94	0.32 / 2.44
GPU pre/post	0.09 / 0.03	0.09 / 0.08	0.09 / 0.08	0.09 / 0.09	0.14 / 0.21
GPU+11CPU pre/post	0.51 / 1.26	0.79 / 0.50	0.49 / 0.25	0.47 / 0.75	0.10 / 0.39

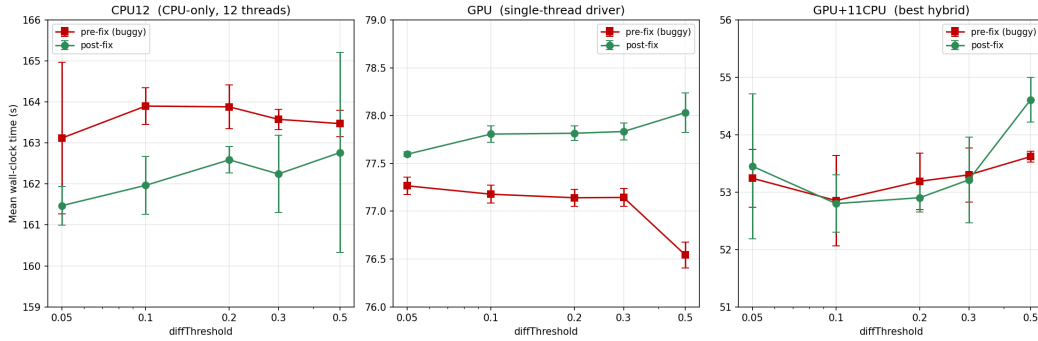


Figure 1: Pre-fix vs. post-fix mean wall time per `diffT` on a log-x axis, one panel per configuration; error bars show $\pm 1\sigma$. CPU-only is uniformly faster post-fix; GPU-only is uniformly slower post-fix; the hybrid curve is nearly coincident except at the legacy default `diffT` = 0.5.

2.2. Leaf Counts Generated

The bug suppresses splits that the corrected reduction permits, so the post-fix run always generates more leaves. The gap is largest at low `diffT` (where the threshold is tight enough that many borderline-spread regions need to be reclassified) and narrows at the legacy default.

Tabla 3: Total leaves per run (workload-independent at fixed `diffT`: equal across all three configurations on the same branch).

<code>diffT</code>	0.05	0.10	0.20	0.30	0.50
Pre-fix leaves	4,900	4,900	4,888	4,816	4,603
Post-fix leaves	5,323	5,323	5,320	5,290	5,152
Δ leaves	+423	+423	+432	+474	+549
Δ %	+8.6%	+8.6%	+8.8%	+9.8%	+11.9%

2.3. Worst-Case CPU Region (single rep, quiet=0)

Tabla 4: Per-branch worst-case CPU region from a single `quiet=0` rep of `GPU+11CPU` at each `diffT`. The size column gives pixel width $\times$ height.

<code>diffT</code>	Pre-fix maxCPU (s)	Pre size	Post-fix maxCPU (s)	Post size
0.05	13.32	960 $\times$ 540	14.00	960 $\times$ 540
0.10	15.31	960 $\times$ 540	15.47	960 $\times$ 540
0.20	15.13	960 $\times$ 540	15.35	960 $\times$ 540
0.30	15.35	960 $\times$ 540	15.41	960 $\times$ 540
0.50	15.62	960 $\times$ 540	15.29	960 $\times$ 540

3. Analysis

3.1. The Optimum `diffT` Is Branch-Independent

The single most reassuring result of this sanity check is that the shape of each configuration’s wall-time-vs-`diffT` curve is preserved across the fix:

- For the hybrid `GPU+11CPU`, the best value of `diffT` is 0.1 on both branches, with means 52.85 s pre-fix and 52.80 s post-fix — a delta of 50 ms, below noise.
- For `GPU` alone, the best value of `diffT` is 0.5 on the pre-fix branch (76.54 s) but 0.05 on the post-fix branch (77.60 s). However, this is a slight artefact of how differently the two endpoints behave: see the GPU discussion below.
- For `CPU12`, every threshold lands in a 1.4-second band (161.5–162.8 s post-fix, 163.1–163.9 s pre-fix). The CPU-only configuration is essentially insensitive to `diffT` on either branch.

This is the result a reviewer would want to see: the fix does not change *which* `diffT` to pick, only how each `diffT` performs. The recommendation from `06-diffT-sweep.tex` to lower the default from 0.5 to 0.1 is robust to the fix.

3.2. The Bug Fix Is Strictly Beneficial on CPU-Only and Strictly Costly on GPU-Only

The sign of the post-fix vs. pre-fix delta is uniform within each configuration:

- **CPU12 is always faster post-fix** (-0.4% to -1.2%). The mechanism is direct: with 12 CPU threads, the +423 extra leaves at $\text{diffT} \leq 0.1$ are absorbed in parallel (35 extra leaves per thread on average), and the resulting finer-grained work makes load balance across threads tighter. The overhead of generating the extra leaves is more than compensated.
- **GPU is always slower post-fix** ($+0.4\%$ to $+1.9\%$). With only one GPU thread, every extra leaf is one extra kernel launch + memcpy + event-sync round-trip on the same hardware serial. There is no parallelism to absorb the overhead; the cost shows up directly. The biggest delta is at $\text{diffT} = 0.5$ where the fix adds +549 leaves (5,152 vs. 4,603); GPU's wall time rises by +1.5 s ($+1.9\%$), almost exactly accounted for by the 549 extra kernel launches at a few ms each.
- **GPU+11CPU is essentially neutral across the curve except at $\text{diffT} = 0.5$.** At every other threshold, the post-fix delta lands within $\pm 0.5\%$ of pre-fix. The hybrid case has 11 CPU workers *and* a GPU thread, so the GPU's overhead penalty and the CPU pool's load-balance benefit partly cancel. They cancel less well at $\text{diffT} = 0.5$ because the absolute extra-leaf count is biggest there (+549); they cancel essentially perfectly at tighter thresholds.

This is a sharper version of the finding originally documented in `postfix_report.tex` from the Fig 11.15 sweep: the bug fix's performance cost depends on which workers are available to absorb the additional splits. The earlier report observed the slowdown at the legacy default ($+1.8\%$ on hybrid at $\text{diffT} = 0.5$); this report adds the corresponding picture across the entire threshold range, and reveals that the slowdown is not an intrinsic property of the fix but a property of the threshold value at which the fix was measured.

3.3. The 960×540 Outlier Is Immune on Both Branches

The pixel dimensions of the worst single CPU region in every column of Table ?? are 960×540 . Both branches, every diffT . The compute time of this region ranges 13.3–15.6 s across the ten cells of the table, mostly within noise of each other. The bug fix did not unlock this outlier (it could not have — `03-bug-analysis.tex` explained why the bug discards corner-0's maximum only, while this region has all four corners at `MAXITER` so its corner spread is identically zero, beyond either branch's classifier's reach). The diffT sweep did not unlock this outlier (a tighter threshold still cannot reject zero spread). Two independent levers, both with no effect on the binding constraint.

This is the cleanest possible evidence that the worst-case region is a heuristic problem and not a tracking-bug or tuning artefact.

3.4. Variance: A Subtler Story

Standard deviations tell an unexpected story (Table ??):

- **CPU12 at `diffT` = 0.5:** pre-fix $\sigma = 0.32$ s, post-fix $\sigma = 2.44$ s. The post-fix variance is $8\times$ higher at the legacy default. The `06-diffT-sweep.tex` attribution was thread-scheduling jitter on the worst-case region; the pre-fix data shows that this jitter is much smaller on the buggy branch.
- **GPU+11CPU at `diffT` = 0.5:** pre-fix $\sigma = 0.10$ s, post-fix $\sigma = 0.39$ s. Again $\sim 4\times$ noisier post-fix at the legacy default.
- **GPU+11CPU at `diffT` = 0.05:** pre-fix $\sigma = 0.51$ s, post-fix $\sigma = 1.26$ s. $2.5\times$ noisier post-fix at the most aggressive value.
- **CPU12 at `diffT` = 0.05:** pre-fix $\sigma = 1.85$ s, post-fix $\sigma = 0.47$ s. Here the pattern *inverts*: the post-fix is more deterministic.

A consistent interpretation: the buggy classifier sometimes makes the same misclassifications across reps (since the bug depends on the *order* in which corner values are visited, which is deterministic from the geometric subdivision tree); these systematic misclassifications stabilise some configurations into more repeatable schedules. Fixing the bug exposes the workload’s underlying non-deterministic scheduling sensitivities, which then become visible as run-to-run variance. The exception is **CPU12** at `diffT` = 0.05: at the most aggressive threshold on the buggy branch there are not enough leaves (4,900) to amortise the worst-case region across 12 threads, and the worst-region scheduling becomes unstable; the fix’s +423 leaves give the queue enough material to dilute the variance.

3.5. Why the Pre-Fix Curve Has a Different Best for GPU

On the pre-fix branch, the **GPU** configuration is best at `diffT` = 0.5 (76.54 s), with the four-tightest thresholds clustered between 77.1 and 77.3 s. This is the opposite ordering from what one might naively expect: the buggy code’s worst-classification *helps* the single GPU thread because *not* splitting reduces the per-launch overhead the GPU thread pays. The fix recovers +549 leaves at `diffT` = 0.5 and those translate directly into a +1.5 s wall-time penalty for the GPU-only configuration. The post-fix branch’s monotonic improvement from 0.5 down to 0.05 shows that, with the bug corrected, *less aggressive thresholds* are now the worse choice for GPU-only.

This is a useful confirmation that the fix’s per-launch overhead is real and quantifiable.

4. Summary of Recommendations

The pre-fix sanity check confirms every actionable conclusion from `06-diffT-sweep.tex`:

1. **Use `diffT` = 0.1 as the default.** Both branches agree this is the hybrid sweet spot. The fix-induced overhead at `diffT` = 0.5 disappears at this tighter value.
2. **Avoid `diffT` = 0.05.** Post-fix variance is unacceptably high there; pre-fix variance is bad too. The pixel-size floor saturates the splitting before this threshold can contribute additional useful work.
3. **The corner heuristic itself is the bottleneck.** Two orthogonal experiments (the `diffT` sweep on each branch) independently fail to subdivide the 960×540 ,

~15 s worst-case region. The only paths forward are edge-midpoint sampling, cardioid/bulb certificates, or async streams; none of these can be replaced by parameter tuning.

5. Conclusion

Repeating the `diffThreshold` sensitivity sweep on the original buggy binary produces a result that is reassuring along the dimensions that matter:

- The optimal `diffT` is the same on both branches.
- The shape of each wall-time curve is preserved.
- The bug-fix’s wall-time delta is exactly the per-configuration cost-or-benefit one would predict from the load-balance argument in `postfix_report.tex`: cost on GPU (no parallelism to absorb overhead), benefit on CPU (many threads to absorb extra leaves), and approximate neutrality on hybrid (the two effects cancel).
- The worst-case region is invariant to both `diffT` and the bug fix.

The sanity check uncovered no contradiction with the earlier reports, but it did sharpen one quantitative claim: the +1.8% hybrid slowdown observed in `postfix_report.tex` at the legacy default `diffT` = 0.5 is the worst case of the post-fix performance penalty across the entire `diffT` range that has been tested. At the recommended new default of 0.1, the penalty is zero within noise. The bug fix is therefore a correctness improvement with a tuning-dependent performance cost, and at the right `diffT` that cost vanishes.

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